Mahendran E

Site Reliability Engineer | Kubernetes | Observability | Incident Response | Multi-Cloud
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SUMMARY

Site Reliability Engineer with 3.5+ years keeping production infrastructure reliable at scale: 1,000+ servers across 30+ environments and 10+ products on AWS, Azure, Oracle Cloud (OCI) and on-premises. Delivered 99.9% uptime across clustered databases and brokers, cut MTTD by 60% with OpenTelemetry, ClickHouse and Grafana observability, and cut RTO by 70% with automated backup and restore. Runs identity-verification and fintech platforms for major UAE banks at 1,000+ requests/second, with 1 TB+ of data across PostgreSQL (Patroni) and 5 ClickHouse clusters, Kafka and Pulsar, and vLLM on NVIDIA GPUs for AI face-matching and document-extraction workloads. Led a 4-engineer team and handled production network and security escalations directly with bank clients.

TECHNICAL SKILLS

- **Observability:** Prometheus, Alertmanager, Grafana, OpenTelemetry, Vector, ClickStack (HyperDX), Elasticsearch
- **Kubernetes & Containers:** Kubernetes, Docker, Helm, ArgoCD (ApplicationSets), Argo Rollouts, Istio Ambient Mesh, Envoy Gateway (Gateway API), Kyverno, kubeconform, kube-score, External Secrets Operator, Cluster Autoscaler, HPA
- **Linux:** RHEL, Oracle Linux, Ubuntu, Amazon Linux; systemd, file systems, package management, performance tuning, server hardening; troubleshooting with tcpdump, mtr, netstat and traceroute
- **Databases & Messaging:** PostgreSQL (Patroni), MongoDB, Redis, etcd, ClickHouse, Milvus, Qdrant, MinIO, Apache Kafka, Apache Pulsar
- **AI Infrastructure:** vLLM, LLM inference, NVIDIA GPU servers on AWS, vector databases (Milvus, Qdrant), AI face-matching and document-extraction pipelines
- **Networking & Security:** TCP/IP, DNS, routing, FortiGate (IPsec VPN, firewall policy), WireGuard, Nginx, Traefik, HAProxy, Consul, dnsmasq, HashiCorp Vault, SSL/TLS certificate lifecycle, mTLS, WAF, NetworkPolicy, Pod Security Standards, PCI DSS
- **Infrastructure as Code:** Terraform (multi-layer remote state, workspaces, import blocks, terraform test), Ansible, SOPS / age
- **Cloud:** AWS (EC2, EKS, VPC, IAM, ELB, RDS, S3, CloudFront, WAF, Lambda, CloudWatch, Security Hub); Azure (AKS, VNet, Virtual Machines, Application Gateway, Blob Storage, Azure DevOps); Oracle Cloud OCI (OKE, VCN, DRG, NSG, Flexible Load Balancer, WAF, Vault, Object Storage, FSS); Cloudflare (DNS, Zero Trust Access, Tunnels, WARP, Gateway)
- **CI/CD & DevSecOps:** GitLab CI/CD (primary), Jenkins (Groovy shared libraries, Job DSL), GitHub Actions, Azure DevOps Pipelines, Bitbucket Pipelines, Nexus, SonarQube, gitleaks, shellcheck, image scanning
- **Languages:** Bash, Python, SQL, YAML
- **Practices:** SRE, SLIs/SLOs, Incident Management, On-call, RCA / Postmortems, Runbooks & SOPs, Disaster Recovery, Capacity Planning, AI-Assisted Engineering (Claude)

PROFESSIONAL EXPERIENCE

Site Reliability Engineer May 2025 – Present
Eurusys (internal transfer from Signzy Technologies) Abu Dhabi, UAE

- Led a 4-engineer DevOps/SRE team operating 1,000+ on-premises and cloud servers across 30+ environments and 10+ identity-verification and fintech products. Owned on-call incident response, RCA reports, runbooks and SOPs, and client-facing network, security and DevOps troubleshooting for major UAE banks.
- **Production incident response:** Resolved production storage outages across MinIO and Milvus clusters (disk exhaustion, disk failure, etcd/MinIO metadata drift), restoring service through MinIO erasure-coded self-healing and a full rebuild of 150M Milvus vectors from ClickHouse. Documented each RCA and turned the fixes into runbooks.
- **Data platform at scale:** Operate 5 ClickHouse clusters and Patroni PostgreSQL holding 1 TB+ of data, plus Kafka and Pulsar messaging, behind products serving 1,000+ requests/second.
- **Observability:** Built Prometheus, Alertmanager and Grafana monitoring, plus log and trace pipelines with OpenTelemetry, Vector, ClickStack (HyperDX), ClickHouse and Elasticsearch. Improved mean time to detect (MTTD) incidents by 60%.
- **Backup and DR:** Built incremental backup and automated restore in Python and Bash for PostgreSQL, MongoDB, ClickHouse and Redis. Cut recovery time objective (RTO) by 70% with near-zero data loss.
- **AI infrastructure:** Run vLLM inference on NVIDIA GPU servers in AWS for AI face-matching and document-extraction workloads in identity verification, with Milvus vector search over 150M+ face embeddings

powering one-to-many face matching.

- **Ansible automation:** Automated end-to-end server lifecycle with Ansible across RHEL, Oracle Linux, Ubuntu and Amazon Linux: baseline setup, production hardening, Docker, and highly available clusters of 20+ stateful tools (Patroni PostgreSQL, MongoDB, Redis, etcd, Kafka, Pulsar, ClickHouse, Milvus, Qdrant, MinIO, Consul, Vault). Achieved 99.9% uptime and cut manual provisioning time by 75%.
- **Kubernetes platform:** Designed and built a multi-tenant Kubernetes platform on AWS EKS, Azure AKS and Oracle OKE: one shared cluster per environment (staging, UAT, production) with per-product node pools, so 10+ interdependent products communicate in-cluster. Implemented Istio ambient mesh (mTLS), Envoy Gateway (Gateway API), External Secrets with cloud vaults and SOPS, per-product HAProxy database proxies and Cluster Autoscaler right-sizing, which cut node compute cost by about 25%.
- **Networking and self-hosted services:** Designed service discovery with Consul, Traefik and dnsmasq, and ran FortiGate site-to-site VPN and firewall troubleshooting. Self-hosted GitLab, Nexus, SonarQube, Listmonk and WireGuard VPN, cutting third-party licensing cost by 40%.
- **Secrets and migration:** Centralized secrets in HashiCorp Vault across 100+ production VMs, removing hardcoded credentials. Migrated Docker-based deployments to Kubernetes with Helm and CI/CD, cutting deployment time by 65%.
- **Deployment CLI with policy gates:** Built a single `k8s` CLI that runs render, kubeconform, Kyverno and kube-score validation, then Helm apply, with unit tests. It blocks misconfigurations before release and reduced new-service onboarding from days to under 2 hours.
- **GitOps:** Moving delivery to ArgoCD (ApplicationSets, a Config Management Plugin that runs the render pipeline) with Git as the single source of truth. Evaluated Argo Rollouts for canary and blue-green releases with metric-based analysis and automatic rollback.
- **Layered Helm architecture:** Standardized all Kubernetes delivery on a 7-tier Helm values hierarchy: a foundation chart (namespaces, quotas, RBAC, NetworkPolicy, Pod Security Standards), per-product microservice charts (HPA, PDB, topology spread, probes, priority classes) and infrastructure charts (Istio, Envoy Gateway, External Secrets, OpenTelemetry, monitoring).
- **Managed Kubernetes:** Ran production EKS, AKS and OKE clusters with private API endpoints, workload identity for External Secrets, Calico network policy and CSI storage, alongside self-hosted database clusters. Ran backup/DR, observability and cost audits that cut monthly cloud spend by about 20%.
- **Multi-cloud landing zones in Terraform:** Architected production hub-and-spoke landing zones on AWS, Azure and OCI from one data-driven Terraform codebase. Each estate is split into about 15 layers with independent remote state and one workspace per environment, driven by YAML as the source of truth with JSON Schema validation and plan-time guardrails.

Software Engineer – DevOps Jun 2023 – Apr 2025
Mitsogo Technologies Inc. Chennai, India

- **GitLab CI/CD to Kubernetes:** Built GitLab CI/CD pipelines that build, scan and deploy Dockerized microservices to Kubernetes on Amazon EKS and to EC2, automating build, test and release across services.
- **Multi-account AWS:** Managed multiple AWS accounts across the company, including IAM, VPC networking and account-level security baselines.
- Ran CloudFront CDN with an Nginx reverse proxy, TLS and AWS WAF. Performed AWS security benchmarking with Security Hub.
- Hardened Dockerfiles and Kubernetes manifests with image scanning and security baselines, meeting 50% of Kubernetes best-practice benchmarks.
- Developed Terraform modules for microservice data infrastructure (RDS, ElastiCache, Amazon MQ, Cognito, CloudWatch). Cut infrastructure deployment time by 80%.

CERTIFICATIONS

- AWS Certified Solutions Architect – Associate
- AWS Certified Developer – Associate
- AWS Certified Cloud Practitioner

EDUCATION

Bachelor of Engineering in Computer Science Aug 2019 – May 2023
Government College of Engineering Srirangam Tamil Nadu, India

PUBLICATIONS

Automated Word App Using DevOps Technology (May 2023), Zenodo, doi.org/10.5281/zenodo.7883225